AI-Based Credit Risk Assessment System

A Machine Learning Approach to Loan Evaluation

The Problem: Manual Loan Assessment

Slow & Labor-Intensive

Traditional credit evaluation requires extensive manual review, consuming hours per application.

Inconsistent Decisions

Human bias and fatigue introduce variability into credit judgments, undermining fairness.

Demand for Speed

Banks urgently need faster, data-driven preliminary assessments to stay competitive.

Project Objectives

Build an ML Classification Model

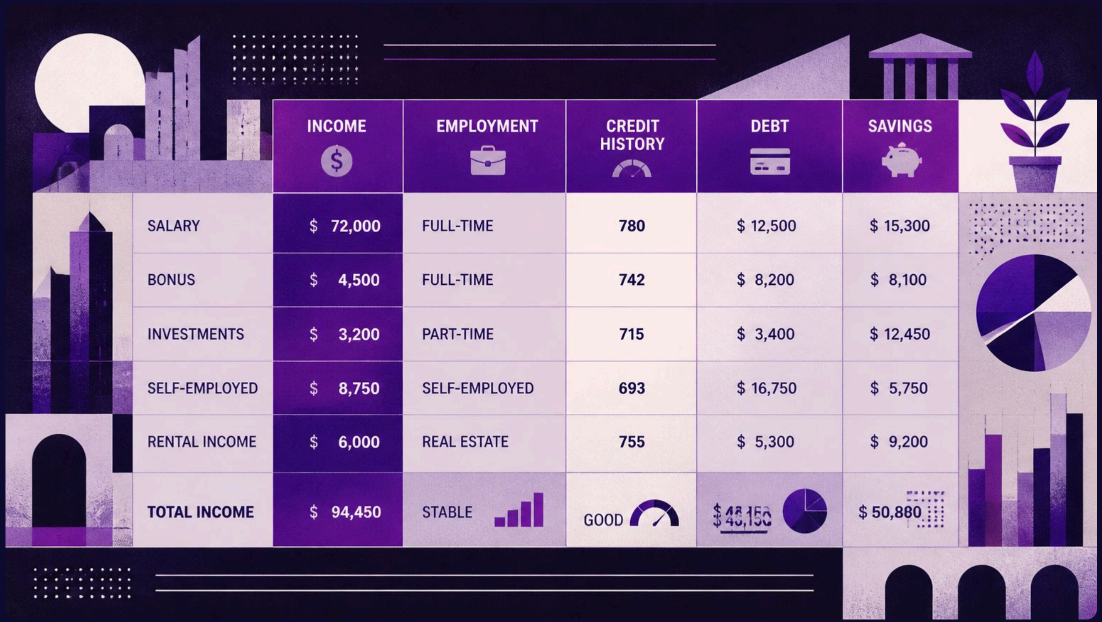
Train a model to classify loan applicants into **Low** or **High** credit risk categories with strong predictive accuracy.

Preprocess & Engineer Features

Clean raw loan application data and derive meaningful financial features to improve model performance.

Deploy an Interactive Web App

Deliver a real-time prediction interface accessible to loan officers and banking analysts.



Dataset Overview

The project uses a structured **Loan Prediction dataset** drawn from historical applicant records.

- **Features:** Applicant income, co-applicant income, employment status, credit history, education level, and property area
- **Target Variable:** Loan approval status — classified as **Low Risk** or **High Risk**
- **Purpose:** Establish a reliable ground-truth basis for supervised learning

Data Preprocessing Pipeline

Handle Missing

Encode Categories

Scale Features

A rigorous preprocessing pipeline ensures the model receives clean, consistent, and normalized inputs — eliminating noise and data quality issues before training begins.

Feature Engineering

Total Income

Applicant Income + Co-applicant Income

Captures the household's combined earning capacity for loan repayment.

Loan Income Ratio

Loan Amount ÷ Total Income

Quantifies the repayment burden relative to total household earnings.

Estimated EMI

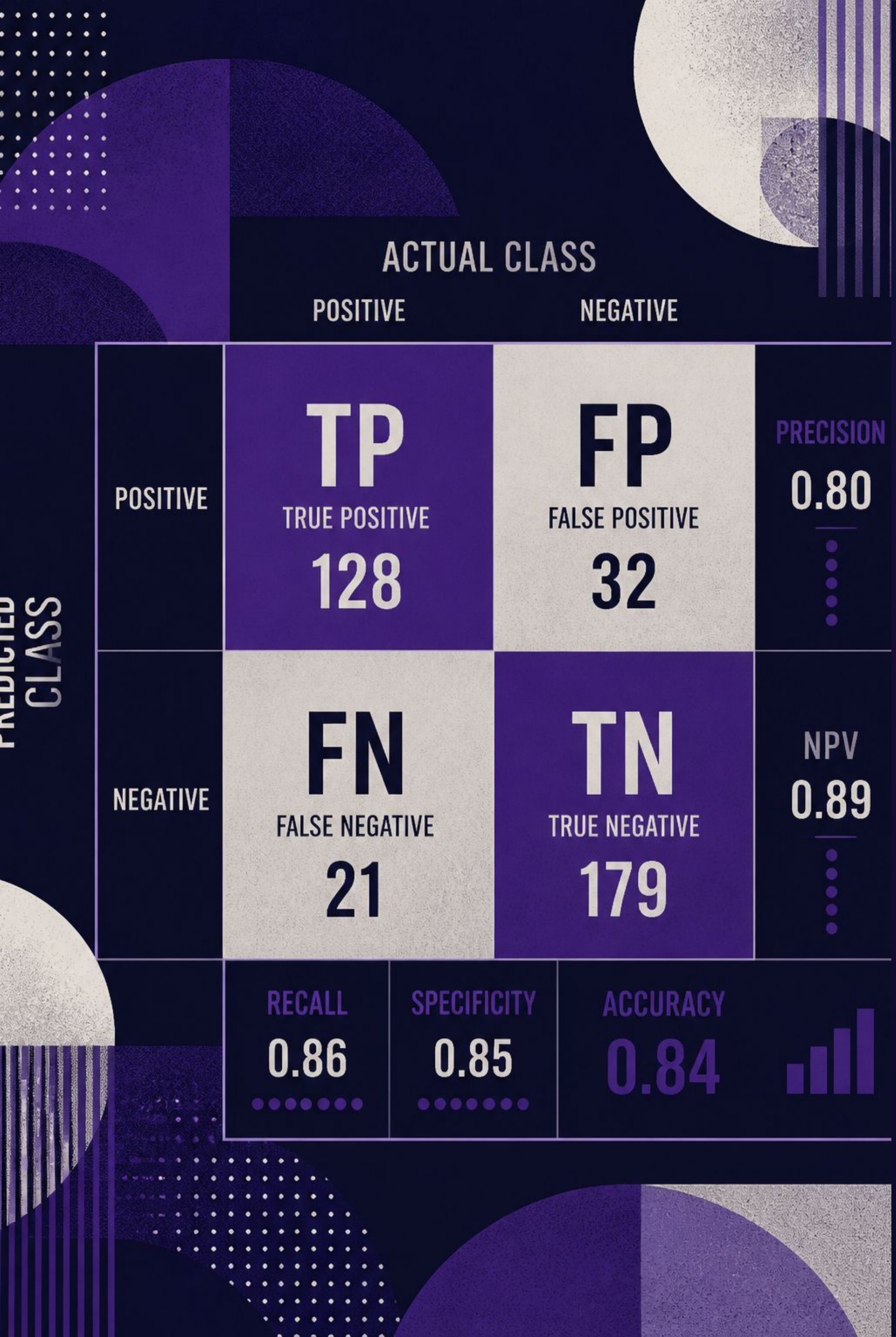
Loan Amount ÷ Loan Term

Approximates the monthly financial obligation imposed on the applicant.

Model Selection: Random Forest

After evaluating multiple algorithms, **Random Forest** was selected as the optimal classifier for this credit risk problem.

- High accuracy and strong generalization through ensemble learning
- Handles both numerical and categorical features without extensive tuning
- Resistant to overfitting via bagging of multiple decision trees
- **80:20 train-test split** used for rigorous performance evaluation



Model Performance

78.86%

Test Accuracy

Achieved on the held-out 20% test dataset

80:20

Train-Test Split

Standard evaluation protocol for generalization testing

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Risk Classes

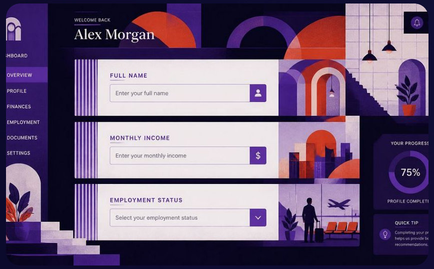
Low Risk and High Risk — evaluated via confusion matrix & classification report

System Architecture

End-to-End Pipeline

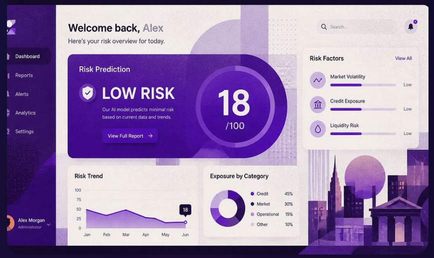
The system integrates seamlessly from trained model artifact to live user interface:

- **Joblib serialization** persists the trained Random Forest model for production reuse
- **Streamlit** powers the interactive front-end with minimal overhead
- **Real-time pipeline** applies identical preprocessing at inference time



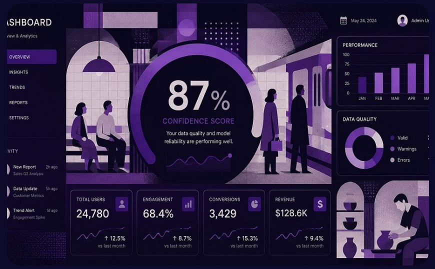
Applicant Input Form

Collects demographics and financial details including income, employment, credit history, and property area.



Instant Risk Prediction

Delivers **Low Risk** or **High Risk** classification in real time, immediately upon form submission.



Confidence Score Display

Shows the model's prediction probability, giving loan officers a quantified measure of reliability.

User Interface Features

Key Advantages

Seconds, Not Hours

Predictions delivered instantly — replacing manual processes that take hours per application.

Consistent & Unbiased

Algorithmic assessments eliminate human bias, ensuring every applicant is evaluated on the same criteria.

Scalable & Reusable

The modular architecture supports future banking applications and can scale to larger datasets with ease.

This system demonstrates that machine learning can meaningfully augment — and accelerate — responsible lending decisions in modern banking.